

Unequal Job Opportunities and Well-Being Inequality

A Latent-Jobs Structural Decomposition

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Observed

C_i

Gini of consumption

single adults 0.2633 couples 0.2268

Well-being

M_i

Gini of the attained-bundle money metric

single adults 0.2498 couples 0.1974

low earnings $\Leftarrow$ tastes **or** opportunities **or** resources

equivalised, household-weighted, within each population

Compensation

unequal job opportunities should not become
unequal well-being

Responsibility

preference-based choices should not be
compensated

Compensation vs. Responsibility

$\nexists W : \text{Full Compensation} \wedge \text{Full Responsibility}$

[Fleurbaey & Maniquet]

How much inequality in money-metric well-being is associated with unequal job opportunities rather than heterogeneous preferences, once labour supply is modelled as choice among latent jobs?

1. Do observed labour-supply choices reflect preferences alone, or also heterogeneous latent job opportunities?
2. How do conclusions differ when welfare evaluates the attained bundle versus the ex-ante opportunity prospect?
3. In a restricted P/A/B decomposition, how much welfare inequality is associated with preferences, local access, and earning opportunities?

Inequality of opportunity

Outcome inequality



circumstances

versus

responsibility-sensitive factors

Typical circumstances

Typical outcomes

parental background · sex · ethnicity or nationality ·
birthplace or region

earnings · income · consumption

What share of observed inequality is associated with circumstances?

Italy
19.5%

Checchi & Peragine 2010

Brazil
10–37%

Bourguignon, Ferreira & Menéndez 2007

France
12.9%

Brunori, Ferreira & Peragine 2013

Different outcomes, circumstance sets and methods.

From circumstances to job opportunities

	Empirical EOp	This paper
Opportunity	observed circumstances or types	estimated latent-job distribution
Outcome	earnings, income, consumption	money-metric well-being
Structural objects	—	preferences · access · earning opportunities
Strategy	compare circumstance-defined distributions	equalise structural components, recompute welfare, decompose inequality

An inequality-of-opportunity question with a structural opportunity object.

Latent jobs

Dagsvik & Jia
Aaberge–Colombino tradition
Capéau, Decoster & Dekkers

Welfare

Fleurbaey & Maniquet
Decoster & Haan
Jacquet, Jia & Thoresen

Decomposition

Shorrocks · Shapley

The contribution is the structural equalisation of estimated preferences, access and earning opportunities followed by recomputation of money-metric welfare inequality.

$j = (e, k, h, w)$

4 occupation groups

$C_i(j)$ priced through EUROMOD

$$v_i(j) = L_i(j) + \beta_c \log\left(\frac{C_i(j)}{\lambda_c}\right)$$

L_i

value of leisure: age; children for
women

β_c

weight on log consumption,
estimated

λ_c

units convention

Opportunities: the choice probability

$$P_i(j) = \frac{\exp\{v_i(j)\} g_i(j)}{\int \exp\{v_i(k)\} g_i(k) d\nu(k)}$$

access

employment mass

hours

bands

occupation

by sex

wage offers

education, experience

valued or plentiful

How are preferences and opportunities separated?

Preferences	smooth systematic utility over hours
Access	local unemployment and geography, excluded from utility
Hours opportunities	band structure
Wage opportunities	education and experience shift the offer distributions
Budgets	EUROMOD generates heterogeneous nonlinear consumption schedules

These restrictions give different empirical variation to preferences and opportunities.

EU-SILC France

collected 2016, incomes 2015

1,540 single-adult households

2,223 couple households

EUROMOD

French 2015 policy system

$h \in [5, 70]$ $w \in [2, 590]$

100 priced alternatives per household

Household-specific opportunities

free coordinates 41

population fit 0.0140

Common opportunities

free coordinates 10 / 16

population fit 0.0273 / 0.0264

criterion worse by at least 141.64

choices reflect opportunities, not preferences alone

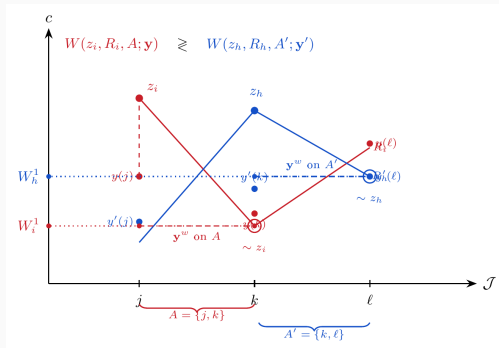
single adults; population fit is the mean absolute deviation of predicted from observed moments

Selected estimates

	Single adults	Couples
β_c consumption weight	2.0387 (0.2917)	2.1017 (0.2939)
$\beta_{E,s}$ local unemployment exposure, in access	-1.4422 (0.2356)	-1.1924 (0.1557)
β_{wH} high education, wage-offer location	0.1491 (0.0308)	0.1817 (0.0197)
σ log-wage dispersion	0.3815 (0.0133)	0.3631 (0.0072)

cluster-robust standard errors in parentheses; the two populations are estimated separately

Welfare: attained outcomes versus job prospects



Own-set equal-consumption equivalents: each individual gets a common consumption on every job in their own ability set; the level at which the preferred job is indifferent to the attained bundle is their money metric, comparable across individuals. Adapted from Haydar and Maniquet (2026), work in progress.

What constant consumption over the household's own job environment is equivalent to its prospect?

$$J_i = \int e^{L_i(j)} \left(\frac{C_i(j)}{\lambda_c} \right)^{\beta_c} \widehat{g}_i(j) d\nu(j), \quad H_i = \int e^{L_i(j)} \widehat{g}_i(j) d\nu(j),$$

$$M_i^{\text{EA}} = \lambda_c \exp \left\{ \frac{\log J_i - \log H_i}{\beta_c} \right\}$$

J_i

the actual prospect

H_i

same jobs, equal pay

M_i^{EA}

flat-consumption equivalent

What is the money equivalent of the attained bundle?

$$M_i^{\text{att}} = C_i^{\text{obs}} \exp \left\{ \frac{L_i(j_i^{\text{obs}}) - L_i(o)}{\beta_c} \right\}$$

o

non-employment, reachable by all

j_i^{obs}

the only route for opportunities

Inequality and Shapley decomposition

P

preferences: age, children

A

local unemployment
exposure, region, urban
or rural location, year

B

wage-offer location

held fixed

household resources,
needs and composition

$$\phi_k^P = \sum_{S \subseteq \{P, A, B\} \setminus \{k\}} \frac{|S|! (3 - |S| - 1)!}{3!} \left[v^P(S \cup \{k\}) - v^P(S) \right]$$

$$v^P(S) = I^P(\emptyset) - I^P(S) \quad I^P: \text{household-weighted Gini of } M^P$$

How much is associated with opportunities?

Ex-ante prospect

equivalised

Single adults

20.3%

Couples

7.9%

Attained bundle

equivalised

raw 14.8%

1.9%

raw 21.3%

3.5%

raw 2.4%

raw 6.7%

A + B as % of the relevant baseline Gini

The two measures also differ in aggregation: the prospect measure inverts within each household; the attained-bundle measure averages inequality across simulated realisations.

Which opportunity channel matters?

Single adults

access
15.5%

earning opportunities

4.8%

$A + B = 20.3\%$

Couples

access
3.7%

earning opportunities

4.2%

$A + B = 7.9\%$

prospective welfare, equivalised: % of baseline Gini

single adults: access > earnings couples: earnings > access

attained-bundle welfare: earnings > access for both populations

Where does baseline inequality go?

shares of baseline Gini · prospect welfare, equivalised

	Single adults	Couples
Preferences ϕ_P/l_0	0.1%	-1.8%
Local access ϕ_A/l_0	15.5%	3.7%
Earning opportunities ϕ_B/l_0	4.8%	4.2%
Other / outside current P-A-B decomposition $1 - (\phi_P + \phi_A + \phi_B)/l_0$	79.6%	93.9%
Total	100.0%	100.0%

Couples: A + B alone = 7.9%; P + A + B = 6.1% because the preference contribution is negative (-1.8%).

Current decomposition equalises P, A and B only.

What is not claimed

Scope

Household resources, needs and composition
held fixed

Access: local access, not total opportunity

Preliminary decomposition

Not claimed

Not causal

No parameter uncertainty yet

Preferences are not responsibility

Neither perspective is designated primary

prospects versus attained outcomes: two different welfare questions

1. Choices reflect opportunities, not preferences alone.
2. Prospects: access leads for single adults. Attained outcomes: earnings lead. Couples: earnings lead under both.
3. Opportunities account for a bounded part of welfare inequality; the preference contribution changes sign with equalisation.

Next: extend the decomposition to household resources and needs.

Appendix map

Literature	EOp benchmarks	Literature		
Model	Utility specification	Opportunity density	Likelihood	Identification
	Observational equivalence			
Estimation	Preferences	Employment access	Hours and occupation	Wage offers
	Common opportunities	Inference	Parameter uncertainty	
Fit	Extensive margin	Fit by margin	Thirty-seven-hour point	Matched households
	Marginal rate of substitution			
Welfare	ATT derivation	EA derivation	EA validation	Equivalisation
	Prospect reference	ATT numerical design	Aggregation orders	Theory scope
Decomposition	Central figure	EA Gini points	ATT Gini points	EA channel shares
	ATT channel shares	Composition within P/A/B	Shapley vs one-factor	Outside P-A-B
	Robustness			
Data	Sample selection	EUROMOD pricing	Numerical design	

Backup: inequality-of-opportunity benchmarks

	share	outcome	data	measure	source
Italy	19.5%	gross earnings	SHIW 1993–2000	MLD; parental education	Checchi & Peragine (2010), p. 445
Brazil, urban	10–37%	male hourly earnings	PNAD 1996	Theil; five circumstances	Bourguignon et al. (2007), abstract
France	12.9%	post-tax earnings	EU-SILC 2005	MLD; five circumstances	Brunori et al. (2013), Table 1

These benchmarks

shares of earnings inequality attributed to observed circumstances

This paper

a share of welfare inequality from estimated latent opportunity components, resources and needs held fixed

Choice sets and restrictions

Aaberge, Colombino & Wennemo (2009): alternative representations of the choice set

Beffy, Blundell, Bozio, Laroque & Tô (2019): labour supply with restricted choices

Latent jobs and welfare

Aaberge, Dagsvik & Strøm;
Dagsvik & Strøm; Dagsvik & Jia;
Capéau, Decoster & Dekkers
Fleurbay & Maniquet; Decoster
& Haan; Bargain et al.; Jacquet,
Jia & Thoresen

Inequality of opportunity and decomposition

Roemer; Checchi & Peragine;
Bourguignon, Ferreira &
Menéndez; Brunori, Ferreira &
Peragine
Shorrocks; Sastre & Trannoy;
Creedy & Hérault; Mühlhan

Backup: observational equivalence

Without restrictions

a taste against certain hours and a scarcity of jobs at those hours predict the same choices

With restrictions

smooth tastes, banded availability and excluded access shifters move different parts of the choice distribution

If availability heterogeneity is omitted, some of its effects can be absorbed by the estimated utility component.

maintained restrictions, not tested · regional variation is not a causal design

Backup: attained-bundle numerical design

Sample

the already-priced estimation
panel; no re-estimation and no
new pricing

Counterfactuals

equalised covariates set to
household-weighted means;
attained bundle simulated by
Gumbel-max with common
random numbers

Checks

replications at fixed estimates; an
independent second seed; exact
Shapley closure

short weekly hours are sparsely represented in the integration panel

Monte Carlo ranges are simulation spreads, not confidence intervals

Backup: two aggregation orders

Attained bundle

Gini within each simulated realisation, then averaged

$$I^{\text{att}}(S) = \frac{1}{R} \sum_r \mathcal{G}_w \left(\{M_{i,r}^{\text{att}}(T_S x_i)\}_i \right)$$

r : simulated realisation of attained bundles, common random numbers across coalitions

Prospect

inverted within each household, then one Gini

$$I^{\text{EA}}(S) = \mathcal{G}_w \left(\{M_i^{\text{EA}}(T_S x_i)\}_i \right)$$

$M_i^{\text{EA}} = \lambda_c \exp\{(\log J_i - \log H_i)/\beta_c\}$, integrated over the household's prospect

the average of realised inequality versus the inequality of household equivalents

Backup: the prospect reference

$$M_i^{\text{EA}} = \left(\int C_i(j)^{\beta_c} \omega_i(j) d\nu(j) \right)^{1/\beta_c}, \quad \omega_i(j) = \frac{e^{L_i(j)} \hat{g}_i(j)}{H_i}$$

Own-opportunity reference

the reference prospect keeps the household's own jobs, availability and preferences, and equalises pay

Weighted generalised mean

the metric is a generalised mean of consumption with exponent β_c , weighted by leisure value and availability

the reference moves with the household's opportunity environment

What the companion theory does and does not cover

Covered

The deterministic own-set equal-consumption construction is characterised axiomatically in the companion paper.

Not covered

The stochastic extension to a prospect is this paper's own step and is not characterised there.

The asymmetry

The attained-bundle measure uses systematic utility only, while the prospect measure's expected-maximum construction makes the taste component relevant to the prospect's value.

Whether that asymmetry is the right normative choice is open.

Backup: parameter uncertainty

Estimated

cluster-robust standard errors for
every estimated coordinate,
clustered on the household

Not yet propagated

estimation uncertainty has not
been carried through either welfare
decomposition

Simulation checks

second seeds and repeat runs
measure numerical noise, not
sampling uncertainty

the decomposition shares are point estimates without confidence intervals

Backup: utility specification

$$v_i(j) = L_i(j) + \beta_c \log\left(\frac{C_i(j)}{\lambda_c}\right), \quad L_i(j) = \beta_\ell^g(\mathbf{x}_i) \mathcal{B}(\tilde{\ell}_i(j); \theta_\ell^g)$$

$$\tilde{\ell}_i(j) = \frac{80 - h}{10}, \quad \mathcal{B}(z; \theta) = \frac{z^\theta - 1}{\theta}$$

$\beta_\ell^g(\mathbf{x}_i)$

intercept, centred age and its
square (units of 10 years)

k_i

children under 20, women only

couples

sum of the two spouses' leisure
terms

Backup: opportunity density

$$g_i(o) = 1, \quad g_i(j) = g_i^E g_i^H(h) g_i^{\text{Occ}}(k) g_i^W(w | k) \quad \text{for market jobs } j \neq o$$

access

g_i^E : local unemployment exposure, region, urban or rural location, year

hours

g_i^H : elevated bands over a residual set

occupation

g_i^{Occ} : sex-specific group masses

wage offers

g_i^W : log-normal, truncated to $[2, 590]$

wage-offer location $\mu_i(k)$: education, potential experience and its square, occupation

Backup: sampled-alternative likelihood

$$V_{ij} = v_i(j) + \log g_i(j) - \log q_{ij}$$

$$\Pr(y \mid \mathcal{C}_i) = \frac{n_y \exp V_{iy}}{\sum_{s \in \mathcal{C}_i} \exp V_{is}}$$

q_{ij}

proposal density, fitted out of fold

$\mathcal{C}_i$

observed job and 100 draws: 101
rows

n_y

multiplicity of a repeated draw

Backup: identification

Functional form

smooth preferences in hours;
banded hours opportunities

Exclusion

local unemployment exposure and
geography shift access, not utility

Independence

wage offers independent of hours,
given occupation

maintained, not tested · not a causal design

Backup: preference parameters

	Single men	Single women	Couple men	Couple women
leisure weight, intercept	8.5199 (3.5480) beta_l0_sm	5.8683 (2.1346) beta_l0_sf	3.9814 (0.7466) beta_l0_m	13.1029 (4.7760) beta_l0_f
leisure weight, age	1.4820 (1.0236) beta_l_age_sm	0.0678 (0.4792) beta_l_age_sf	-0.0080 (0.0287) beta_l_age_m	-0.1228 (0.1260) beta_l_age_f
leisure weight, age squared	0.7783 (0.7732) beta_l_age2_sm	1.0000 (at bound) beta_l_age2_sf	0.0065 (0.0030) beta_l_age2_m	0.0073 (0.0103) beta_l_age2_f
leisure weight, children	—	0.1666 (0.4422) beta_l_nkids_sf	—	-0.3055 (1.1472) beta_l_nkids_f
leisure curvature θ_ℓ	-1.6263 (0.3301) theta_l_sm	-0.9274 (0.2161) theta_l_sf	-0.9761 (0.1357) theta_l_m	-1.6863 (0.2458) theta_l_f
consumption weight β_c	2.0387 (0.2917) beta_c		2.1017 (0.2939) beta_c	

cluster-robust standard errors; parameter identifiers in grey; the male children term in couples is structurally zero

Backup: employment-access parameters

	identifier	Singles	Couples
employment constant	beta_E; beta_E_m, beta_E_f	−3.1735 (0.4053)	men −2.1233 (0.3195) women −2.8148 (0.3021)
local unemployment exposure	beta_E_gsur	−1.4422 (0.2356)	−1.1924 (0.1557)
densely populated	beta_E_drgur	−0.0288 (0.2209)	−0.1823 (0.1657)
intermediate density	beta_E_drgmd	0.0641 (0.2614)	−0.4174 (0.1861)
Region 2	beta_E_drgn2	−0.3780 (0.3292)	−0.1546 (0.2442)
Region 3	beta_E_drgn3	−0.1117 (0.3891)	0.0873 (0.2808)
Region 4	beta_E_drgn4	−0.8218 (0.3810)	0.0150 (0.3068)
Region 5	beta_E_drgn5	−0.5162 (0.3274)	−0.1455 (0.2555)
Region 6	beta_E_drgn6	−0.7222 (0.3528)	−0.2844 (0.2748)
Region 7	beta_E_drgn7	−0.5374 (0.3503)	−0.1396 (0.2651)
Region 8	beta_E_drgn8	−0.4512 (0.3401)	−0.1659 (0.2660)

references: the first region and thinly populated areas; couples share the shifters across spouses

Backup: hours and occupation opportunities

Hours bands					
	Singles		Couple men		Couple women
part-time lower	0.0298		−1.1094		−0.5173
	beta_h_pt1		beta_h_pt1_m		beta_h_pt1_f
part-time upper	0.7080		0.0835		1.1586
	beta_h_pt2		beta_h_pt2_m		beta_h_pt2_f
narrow full-time	2.0656		2.2888		2.0284
	beta_h_f35		beta_h_f35_m		beta_h_f35_f
full-time upper	1.9387		2.3801		1.6825
	beta_h_ft		beta_h_ft_m		beta_h_ft_f
long hours	−0.0834		0.6977		−0.3109
	beta_h_lh		beta_h_lh_m		beta_h_lh_f
Occupation masses					
	Singles		Couples		
	men	women	men	women	
Occupation 2	−1.3737	0.1135	−1.5097	0.2074	
	beta_occ_2_m	beta_occ_2_f	beta_occ_2_m	beta_occ_2_f	
Occupation 3	−2.1096	−0.4060	−2.2532	−0.1804	
	beta_occ_3_m	beta_occ_3_f	beta_occ_3_m	beta_occ_3_f	
Occupation 4	−0.3259	0.5632	0.1594	0.8159	
	beta_occ_4_m	beta_occ_4_f	beta_occ_4_m	beta_occ_4_f	

estimates with identifiers; standard errors in the paper

Backup: wage-offer equation

	identifier	Singles	Couples
intercept	beta_w0	2.0138 (0.0561)	2.0480 (0.0335)
low education	beta_w_educL	0.0566 (0.0372)	−0.0433 (0.0217)
high education	beta_w_educH	0.1491 (0.0308)	0.1817 (0.0197)
potential experience	beta_w_pexp	0.2408 (0.0862)	0.5643 (0.0546)
experience squared	beta_w_pexp2	−0.0282 (0.0390)	−0.1598 (0.0243)
Occupation 2	delta_occ_2	−0.0347 (0.0399)	−0.0740 (0.0231)
Occupation 3	delta_occ_3	0.0622 (0.0383)	0.0394 (0.0228)
Occupation 4	delta_occ_4	0.2785 (0.0368)	0.2146 (0.0223)
log-wage dispersion σ	sigma	0.3815 (0.0133)	0.3631 (0.0072)

medium education is the reference; experience in units of 20 years

Backup: common versus household-specific opportunities

Single adults	free coordinates	criterion	population fit
household-specific opportunities	41	6253.463	0.0140
common opportunity distribution	10	6403.974	0.0273
common shape; employment and hours in utility	16	6395.108	0.0264

same households, sampled alternatives and criterion; not converted into a formal test

Single adults

free coordinates 41, interior 40

criterion 6253.463

smallest Hessian eigenvalue 0.1669

Couples

free coordinates 47, interior 47

criterion 10283.034

smallest Hessian eigenvalue 0.0771

cluster-robust standard errors on the household · eigenvalues on the interior block

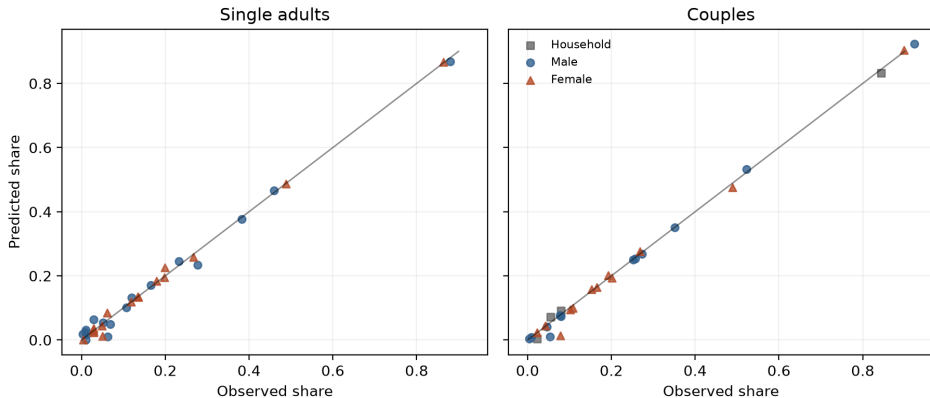
Backup: extensive-margin fit

	weighted accuracy	model-simulated range	integration ratio
single women	85.7%	80.0–85.9%	0.096
coupled women	89.8%	88.1–91.2%	0.049
single men	withheld	—	0.361
coupled men	withheld	—	0.353

withheld where numerical integration error is too large relative to sampling variation

Backup: fit by margin

Corrected population moments: observed versus predicted



Backup: the thirty-seven-hour point

single men

5.2

pp

single women

3.8

pp

coupled men

4.4

pp

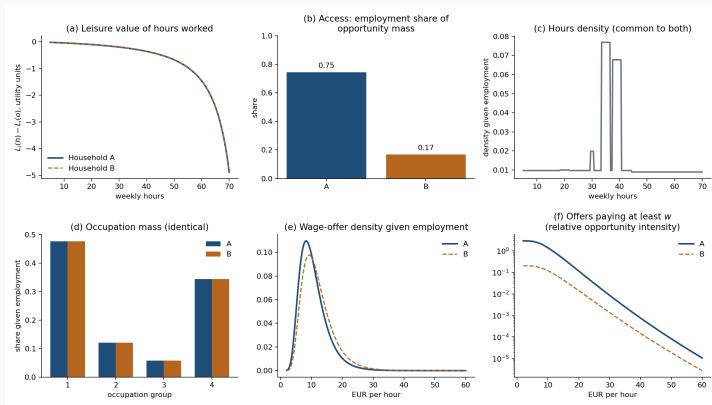
coupled women

6.5

pp

observed minus predicted share at the observed thirty-seven-hour mass point

Backup: matched households

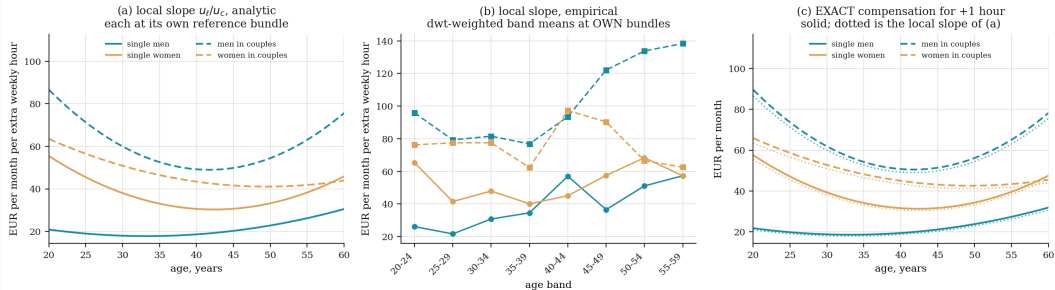


access: $A = 14.5 \times B$

wage-offer location: $B + 11.5$ log points

Backup: marginal rate of substitution

The local compensation u_l/u_c by age and sex, and the exact compensation for one more weekly hour



Backup: attained-bundle derivation

$$L_i(o) + \beta_c \log\left(\frac{M_i^{\text{att}}}{\lambda_c}\right) = L_i(j_i^{\text{obs}}) + \beta_c \log\left(\frac{C_i^{\text{obs}}}{\lambda_c}\right)$$

$$\implies M_i^{\text{att}} = C_i^{\text{obs}} \exp\left\{\frac{L_i(j_i^{\text{obs}}) - L_i(o)}{\beta_c}\right\}$$

Reference job

the preferred job when every job pays the same:
here non-employment for every household

Staying-home equivalent

a verified property of this specification, not a
general theorem

Backup: ex-ante derivation

$$J_i = \int_{\Omega_i} e^{L_i(j)} \left(\frac{C_i(j)}{\lambda_c} \right)^{\beta_c} \widehat{g}_i(j) d\nu(j)$$

$$J_i^{\text{ref}}(m) = \left(\frac{m}{\lambda_c} \right)^{\beta_c} H_i, \quad H_i = \int_{\Omega_i} e^{L_i(j)} \widehat{g}_i(j) d\nu(j)$$

$$J_i^{\text{ref}}(M_i^{\text{EA}}) = J_i \quad \implies \quad M_i^{\text{EA}} = \lambda_c \exp \left\{ \frac{\log J_i - \log H_i}{\beta_c} \right\}$$

a common rescaling of $\widehat{g}_i$ cancels $\cdot \log J_i$: expected utility of the best reachable job, up to a constant

Backup: ex-ante numerical validation

Identities

constant-consumption identity

one-state identity

common intensity-rescaling invariance

flat-reference inversion

exact Shapley adding-up

Numerics

exact reference integral on the priced set

effective sample size and weight concentration

two-seed stability of welfare, Gini and Shapley

finite positive welfare for every household

independent reimplementations

all passed for both populations · a correct computation, not causal identification or parameter uncertainty

Backup: equivalisation

$$M_i^{p,eq} = \frac{M_i^p}{e_i}, \quad e_i : \text{modified-OECD scale}$$

Couples, ex ante: A + B

21.3% $\longrightarrow$ 7.9%

raw $\rightarrow$ equivalised; % of baseline Gini

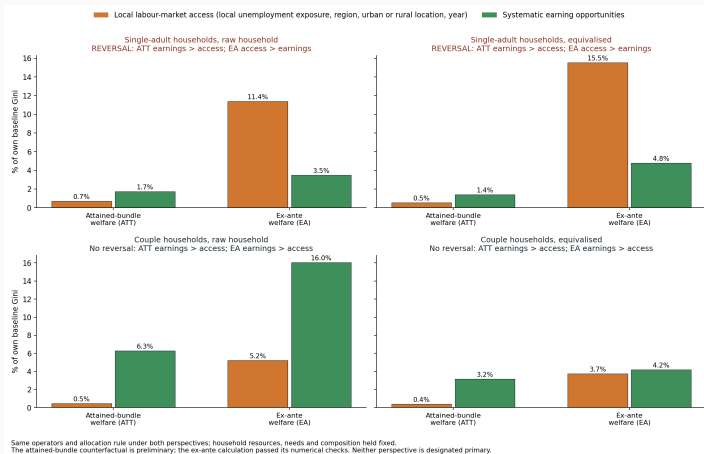
Couples, ex ante: preferences

0.00291 $\longrightarrow$ -0.00212

Gini points, raw $\rightarrow$ equivalised

a normative convention, reported both ways; populations never combined or compared

Backup: central figure



Backup: ex-ante decomposition in Gini points

	scale	baseline Gini	P	A	B	A + B, % of baseline	ordering
Single adults	raw	0.1595	0.00157	0.01812	0.00555	14.8	access > earnings
	equivalised	0.1406	0.00020	0.02183	0.00673	20.3	access > earnings
Couples	raw	0.1277	0.00291	0.00667	0.02049	21.3	earnings > access
	equivalised	0.1200	-0.00212	0.00449	0.00504	7.9	earnings > access

exact Shapley contributions in Gini points; household-weighted

Backup: attained-bundle decomposition in Gini points

	scale	baseline Gini	P	A	B	A + B, % of baseline	ordering
Single adults	raw	0.2337	0.0012	0.0016	0.0040	2.4	earnings > access
	equivalised	0.2451	-0.0003	0.0013	0.0034	1.9	earnings > access
Couples	raw	0.2035	0.0064	0.0009	0.0128	6.7	earnings > access
	equivalised	0.1971	-0.0019	0.0007	0.0062	3.5	earnings > access

exact Shapley contributions in Gini points; preliminary; household-weighted

Backup: ex-ante channels, % of baseline

		P				A		B	
	scale	I_0	Gini pts	% of I_0	Gini pts	% of I_0	Gini pts	% of I_0	
Single adults	raw	0.1595	0.00157	0.99	0.01812	11.37	0.00555	3.48	
	equivalised	0.1406	0.00020	0.14	0.02183	15.52	0.00673	4.79	
Couples	raw	0.1277	0.00291	2.28	0.00667	5.22	0.02049	16.04	
	equivalised	0.1200	−0.00212	−1.76	0.00449	3.75	0.00504	4.20	

each contribution divided by its own baseline Gini I_0 ; signed; the rows are not forced to sum to one hundred, because the remainder is outside the current P-A-B decomposition

Backup: attained-bundle channels, % of baseline

		P			A		B	
	scale	I_0	Gini pts	% of I_0	Gini pts	% of I_0	Gini pts	% of I_0
Single adults	raw	0.2337	0.0012	0.53	0.0016	0.69	0.0040	1.73
	equivalised	0.2451	-0.0003	-0.13	0.0013	0.54	0.0034	1.40
Couples	raw	0.2035	0.0064	3.16	0.0009	0.45	0.0128	6.28
	equivalised	0.1971	-0.0019	-0.98	0.0007	0.37	0.0062	3.15

each contribution divided by its own baseline Gini I_0 ; signed; preliminary

Backup: composition of the explained component

$$s_k = \frac{\phi_k}{\phi_P + \phi_A + \phi_B}, \quad k \in \{P, A, B\}$$

ex-ante prospect, equivalised	Single adults	Couples
Preferences s_P	0.7%	−28.5%
Local access s_A	75.9%	60.6%
Earning opportunities s_B	23.4%	67.9%
Total	100.0%	100.0%

These are shares of the currently explained P/A/B change, not shares of total inequality. Negative components are possible under Shapley attribution.

Backup: Shapley versus one-factor equalisation

$$\text{one-factor: } v^P(\{k\}) = I^P(\emptyset) - I^P(\{k\})$$

$$\text{Shapley: } \phi_k^P = \text{average over all orders of } v^P(S \cup \{k\}) - v^P(S)$$

$$\phi_P^P + \phi_A^P + \phi_B^P = I^P(\emptyset) - I^P(\{P, A, B\})$$

one-factor effects need not add up · Shapley contributions do, exactly

Backup: outside the current P-A-B decomposition

Held fixed

household resources, needs and
composition

Retained

sex-specific and
household-type-specific preference
and opportunity blocks

Not equalised

wage-draw luck and the common
offer spread

wage-offer location differences: at most about 0.10 log points

common offer spread: about 0.37 log points

next: extend the decomposition to household resources and needs, after fixing the appropriate operator

Backup: robustness of the attained-bundle decomposition

Second simulation seed

every coalition Gini and contribution
reproduces closely, including the preference
sign change

Observed job in the simulation

attained under every coalition by 2.9% of single
adults and 3.9% of couples
excluding it moves the total change by at most
9.3%, with no sign change

simulation checks, not confidence intervals

Backup: sample selection

screen	single-adult	couple
the raw France 2016 input file	4,038	5,965
one- or two-adult households (composition screen)	4,038	5,965
every adult aged 20 to 60	2,131	3,662
no adult still in education	2,036	3,521
no old-age, disability or survivor pension receipt	1,755	3,218
labour status in scope	1,598	2,412
no other employable or earning member	1,564	2,323
hours and wage inside the modelled support	1,555	2,275
hours outside [5,70] or unsupported military occupation (ISCO 0) on an employed decider	1,543	2,223
observed chosen alternative priced at non-positive disposable consumption	1,540	2,223
estimation sample	1,540	2,223

Backup: EUROMOD pricing

Inputs

a work arrangement: hours, hourly wage, earnings split; household roster and non-labour inputs

EUROMOD

French 2015 policy system; only deciders receive counterfactual overrides

Output

$C_i(j)$: household disposable income, summed over all members

every sampled alternative and every welfare integration point is priced
tax-benefit accounting identity holds to machine precision

Estimation

101

rows per household: the observed
job and 100 sampled alternatives

Predictive fit

2,048

common integration points per
household

Supports

[5, 70]

weekly hours; wages in [2, 590] per
hour

sampling devices are computational: they are not offers and enter no welfare measure